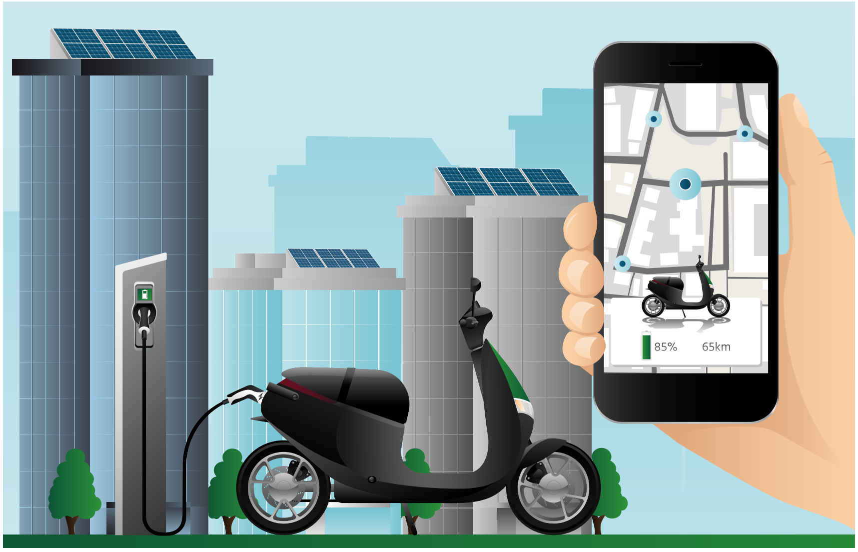

Imran Mulla, head of R&D at Classic Legends (JAWA Motorcycles, Yezdi, BSA), elaborates on the term connected mobility. He says, humans have always had the need to travel, and thus the need for the mode of travel and its accessibility. This accessibility has always been a concern, especially in a country like India. Today, in the era of digital transformation, we heavily rely on a digital ecosystem. "It is the mobile phone that has actually opened the gateway for connectivity," he says.

The onset of the pandemic played a crucial role in quicker adoption and fast-paced development of the EVs and connected mobility. Bob Paul Raj, Head - Connected & Digital Platforms, Mahindra & Mahindra, opines, "The rapid adoption of digital platforms and initiatives, especially during the onset of the pandemic, has exponentially grown, and some key components or growth areas in digital era include process optimisation, better engagement with customers, and thriving new business models. And connected platform is a key milestone on this digital journey for an OEM." Bob believes that all this machine data integrating with context provides value to the entire ecosystem and visibility to the entire business value chain, keeping in mind the growing technology together is what connected mobility means.

Connected mobility for an electric future

The electric vehicle industry in India is growing by leaps and bounds, and the government has taken various steps to promote electric mobility in the country. Despite that, there is still hesitation among vehicle owners to shift to electric mobility.

The reasons for this hesitation are many, including lack of charging infrastructure, higher cost of EVs, and range anxiety. Now, with the introduction of connected mobility, Imran believes that it can provide the EV industry a significant boost such that it gains an edge over other forms of mobility. OEMs are now focusing on making EVs accessible to a larger audience by removing some barriers.

For greater convenience to customers, Imran suggests that an integrated way of transportation has to be taken where safety and user experience are the primary focus areas. He adds, there are multiple fronts to implement this integration, but simply said, it has to be intelligent, integrated, and safe, and with a strong carbon footprint which we are all working towards.

With these fast changes with respect to connectivity, OEMs also need to adapt as it is very new for them. Rane says, "It is difficult for older automotive players to understand how this system works." This is because everything from data transfer to analysis of data works at a very fast rate. The best approach has to be analysed to create and maintain such integration. Further, connectivity brings challenges in terms of backend infrastructure, which needs to be designed and maintained. Maintenance of such sophisticated backend infrastructure is a recurring cost. Will OEMs bear this cost, or will end users bear this cost? All these questions need to be addressed keeping convenience of the consumers in mind.

Connectivity in E2Ws

The rapidly growing number of electric two-wheelers (E2Ws) has taken the market by storm and these are on the way to become IoT devices on wheels. There have been various advancements in battery technologies and power devices, which has enabled the development of a connected space.

OEMs are trying to integrate features like satellite based positioning receiver, cellular modem for cloud connectivity, and a short-range Bluetooth or Wi-Fi link for local tasks. This connectivity enhances user experience and safety, making the overall solution adoptable. "Today, having a small hardware that talks to a CAN is an easy way of making the motorcycle connected," Imran says.

With this connectivity, users will have access to real-time information about their motorcycle, like its geolocation, traffic data nearby, etc. There is a focus on implementing advanced warning systems or early warning systems where a particular congestion on the road is notified much earlier. Another aspect is how personal information like the geoposition of the user is shared among other users for more responsiveness. All of these are the challenges that are being worked upon.

"But to begin with, it is more about providing the convenience and ease to a rider by having diagnostics available which are real-time," he concludes.

Connectivity in agriculture

Connectivity plays a major role in almost every sector by providing the capability to transfer data between the sensors, computers/processors, and the end users. The agricultural sector can also leverage this connected approach, says Bob.

According to Bob, connectivity applies to every stage of farming. There are dedicated tools and algorithms for each stage in farming. Precision agriculture methods are constantly trying to increase crop yields and profitability while lowering the levels of traditional inputs needed to